Figure 3: The Electrum Bitcoin client for Android runs with Python on Android, and provides all features without downloading the block-chain

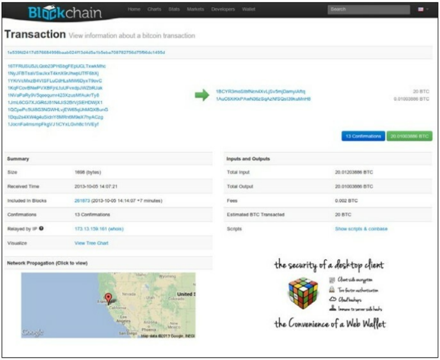

Figure 4: Viewing the details about a Bitcoin transaction on the website—*www.blockchain.info*

obtained from the websites of these clients. The official Bitcoin website list links to the websites of these clients.

How can Bitcoins be used?

Bitcoins are accepted in much fewer places than normal currency and conventional payment methods. Online hosting and other digital goods are some of the limited things that you can buy with Bitcoins. In the US, hotels and restaurants have started accepting Bitcoins; however, such popularity is yet to reach India. A few days back, Silk Road, the secret online market place was taken down by the FBI. It was one of those places which allowed you to buy and sell *anything* with Bitcoin. It was literally a place where people could sell anything, including drugs and weapons. It is estimated that Silk Road had done an estimated business of more than 9.5 million Bitcoins out of the total 11 million Bitcoins in circulation today. What is surprising is that the closure did not have any significant effect on Bitcoin's price. This obviously means that people consider it seriously and it is more than just a medium for black marketeers or a method for money laundering. A lot of online merchants today do not even allow you to buy Bitcoins without extensive verification of your documents and identity. They also have an upper limit beyond which you cannot buy any more Bitcoins. Today, Bitcoin is mainly used by many people for trading and investment purposes as they expect its value to only rise over the next few years.

Bitcoin mining

Bitcoin mining is a resource intensive process and is done only on high performance machines with significant GPU and CPU processing power. Some of the methods of mining Bitcoins are CPU mining, GPU mining, FPGA mining and ASIC mining. The last one is the latest and most efficient technology to mine Bitcoins with the least amount of processing power. Mining takes up so much of resources that it is done by many people in a single pool or group, which works towards a common goal of producing a block that fetches them a total reward of around 50 Bitcoins for their efforts. This reward is to be reduced over time to 25 Bitcoins, then to 12.5 Bitcoins, and so on. Eventually, the miners are expected to be rewarded for their efforts with the small transaction fees that is attached to each transaction. This results in the creation of more Bitcoins, which is why it is known as 'mining' as it resembles the production of similar limited resources like gold and currency.

Explaining the actual processing that happens during mining requires a more detailed understanding of cryptography and hashing. But, in simple words, it is a process that is intentionally made difficult so that blocks are produced at a rate of around one block every ten minutes for all the miners, which is a total of six blocks per hour.

What actually happens during mining is that miners try to produce the next block by using the hash value of the current block, along with the representation of the next set of transactions shown as a Merkle root, which is added along with a nonce value. The nonce value keeps increasing until the correct solution is produced, which is lower than the target value. Whichever miner solves the problem in the fastest time gets the reward of the Bitcoins, and this process keeps repeating.